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Roll no.

Degree: B.TECH Semester: 6th ECAM
END-SEMESTER EXAMINATION, APRIL-MAY, 2025

Course Title: Optimization Techniques in AI & ML
Course Code: EAEPE21

Duration: 03 Hours

Max. Marks: 40

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	What is mathematical Optimization? What is its utility? Write its generic form.	4	CO1
1b	Is there any best or universal optimization algorithm? Justify your statement.	4	CO1
1c	How are Optimization Algorithms classified according to the principle of a biological and physical-based algorithm. How is Backtracking search algorithm-BSA different from Simulated annealing algorithm-SA.	4	CO1
Q 2	Attempt any 2 parts of the following		
2a	What is a metaheuristic algorithm? Explain the importance of exploration and exploitation in optimization.	4	CO2
2b	Explain the concept of evolutionary operators with reference to nature-inspired algorithms.	4	CO2
2c	Explain the working principle of the Simulated Annealing algorithm. How does the concept of temperature help in avoiding local optima? Illustrate with an example.	4	CO2
Q 3	Attempt any 2 parts of the following		
3a	Define and compare the following types of selection methods in GA: (a) Roulette Wheel Selection (b) Tournament Selection	4	CO3
3b	What do you understand by convergence in GA? What are the indicators of premature convergence, and how can it be avoided?	4	CO3
3c	Discuss different variants of GA. What is the role of genetic operators in different variants. Compare with an example.	4	CO2,3
Q 4	Attempt any 2 parts of the following		
4a	What is Fuzzy-Aided PSO? How does it differ from traditional PSO?	4	CO4
4b	Use Particle Swarm Optimization (PSO) to minimize the function: $f(x) = x^2 + 6x + 8$. Perform 2 iterations using 5 particles.	4	CO4,5
4c	Discuss the main components of ACO: pheromone trail, heuristic information, and pheromone update.	4	CO4
Q 5	Attempt any 2 parts of the following		
5a	What is Pareto optimality? Explain how it is used in multi-objective optimization.	4	CO5
5b	Describe the Weighted Sum Method and the E-Constraint Method used in multi-objective optimization	4	CO5
5c	What is NSGA-II? Explain how it solves multi-objective optimization problems using metaheuristic approaches.	4	CO5